Revisiting Bandlimited Signal Formats for GNSS: Appendix

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I. Introduction

A. Scope of the document

The scope of this document is to provide a full derivation of the parameters used in [1]. To do so, let us assume that a global navigation satellite system (GNSS) signal $x(t)$ (at the output of the radio frequency (RF) filter, as outlined in [1, Figure 4]) is received embedded in additive white Gaussian noise (AWGN) within a certain observation time window $[0, T_{\mathsf{obs}}]$, whose extension, typically an integer multiple of the spreading sequence period, is much larger than the chip time T_c.

The variance of any estimation algorithm implemented by the GNSS receiver to derive an estimate of the signal time-of-flight (delay) τ is lower bounded by the *modified CRB (MCRB)* [2] (also known as *average CRB*)

$$\mathsf{MCRB}\,(\tau) = \frac{1}{8\pi^2\beta_x^2 \cdot \mathsf{SNR}}, \tag{1}$$

where $\mathsf{SNR} = P \cdot T_{\mathsf{obs}}/N_0$ is the estimation signal-to-noise ratio (SNR) across the coherent observation window. In the SNR, N_0 is the noise power spectral density (PSD), and the received signal power P, is computed as

$$P = \int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} \mathcal{P}_x(f) df, \tag{2}$$

where $\mathcal{P}_x(f)$ denotes the PSD of the transmitted signal $x(t)$. In (1), β_x is the root mean square (rms) bandwidth of the GNSS signal, also called the Gabor bandwidth (GB) of $x(t)$:

$$\beta_x \triangleq \sqrt{\frac{1}{P}\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} f^2\,\mathcal{P}_x(f) df}. \tag{3}$$

In (2)-(3), $\mathcal{P}_x(f)$ is derived assuming that the ranging code chips are random. The *modified* appellation in the bound's name indicates that it is calculated under this same assumption: specifically, it considers the code as a vector of random, independent and identically distributed (iid) *nuisance parameters* [2], [3]. It is seen that both (2) and (3) assume that the receiver processes the GNSS signal within a specific receiver RF bandwidth B_{rx}: as already mentioned, in case the GNSS signal is strictly bandlimited (i.e., it shows a finite RF bandwidth B_x), we assume $B_{\mathsf{rx}} \geq B_x$ (additional considerations along this line will be done later on) and the bound does *not* depend on B_{rx} (it is only dependent on B_x) as the transmitted signal can be fully processed. This corresponds to the practical case when, to achieve higher spectral efficiency and limit out of band emissions, the bandlimiting shall take place at the satellite navigation signal generation unit (NSGU). When we consider instead (theoretically) band-unlimited signals, the received power and the GB do depend on B_{rx}, hence the bound decreases in general as B_{rx} increases, since more and more of the signal spectrum can be processed.

B. Structure of the document

The remainder of the document is structured as follows. After this introduction, Sect. II reports some useful integrals and notation used throughout this technical note. Sect. III investigates the properties of the non-return-to-zero (NRZ) signal, whereas Sects. IV and V analyze the binary offset carrier (BOC) counterparts. Sects. VI and VII tackle the square-root raised cosine (SRRC) signals (without and with introduction of subcarriers). Sects. VIII and IX address the Gaussian and the Gaussian minimum shift keying (GMSK) formats, respectively, whereas Sect. X considers the orthogonal frequency division multiplexing (OFDM) format.

II. Useful results

$$\int_0^x \sin^2(\alpha)\, d\alpha = \frac{x}{2} - \frac{\sin(2x)}{4} \tag{4}$$

$$\int_0^x \cos^2(\alpha)\, d\alpha = \frac{x}{2} + \frac{1}{4}\sin(2x) \tag{5}$$

$$\int_0^x \frac{\sin^2(\alpha)}{\alpha^2}\, d\alpha = \mathrm{Si}(2x) - \frac{\sin^2(x)}{x} \tag{6}$$

$$\int \frac{\sin^2(\alpha)}{\alpha}\, d\alpha = \frac{\log\alpha}{2} - \frac{\mathrm{Ci}(2\alpha)}{2} \tag{7}$$

$$\int_0^x \alpha\cos^2(\alpha)\, d\alpha = \frac{x^2}{4} + \frac{1}{4}x\sin(2x) + \frac{1}{8}\left[\cos(2x) - 1\right] \tag{8}$$

$$\int_0^x \alpha^2\cos^2(\alpha)\, d\alpha = \frac{x^3}{6} + \frac{1}{8}\left(2x^2 - 1\right)\sin(2x) + \frac{1}{4}x\cos(2x) \tag{9}$$

$$\int_0^{+\infty} \alpha^2 \exp\left\{-\alpha^2\right\} d\alpha = \frac{\sqrt{\pi}}{4} \tag{10}$$

where

$$\mathrm{Si}(z) \triangleq \int_0^z \frac{\sin(y)}{y} dy, \tag{11}$$

is the integral sine function, and

$$\mathrm{Ci}(z) \triangleq -\int_z^{+\infty} \frac{\cos(y)}{y} dy \tag{12}$$

is the integral cosine function.

III. NRZ

The simplest example of a GNSS signal, assuming no data modulation and unit-power, is expressed by

$$x_{\mathsf{N}}(t) = \sum_{k=-\infty}^{\infty} c[k]\, p_{\mathsf{N}}(t - kT_{\mathsf{N}}), \tag{13}$$

where $c[k]$ is the k-th chip of the ranging code, T_{N} is the NRZ chip time, i.e., the inverse of the chiprate R_{N}, and $p_{\mathsf{N}}(t)$ is a unit-amplitude, T_{N}-wide rectangular pulse[1] having energy T_{N}. The corresponding (unit-power) PSD is

$$\mathcal{P}_{\mathsf{N}}(f) = T_{\mathsf{N}} \left[\frac{\sin(\pi f T_{\mathsf{N}})}{\pi f T_{\mathsf{N}}}\right]^2 = T_{\mathsf{N}} \,\mathrm{sinc}^2(f T_{\mathsf{N}}). \tag{14}$$

To drive (3), let us use the definition of *normalized* RF receiver bandwidth $W_{\mathsf{N}} = B_{\mathsf{rx}} T_{\mathsf{N}}$, and compute separately the numerator

$$\begin{aligned}\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} f^2 \mathcal{P}_{\mathsf{N}}(f)\, df &= 2T_{\mathsf{N}} \int_0^{B_{\mathsf{rx}}/2} \frac{\sin^2(\pi f T_{\mathsf{N}})}{\pi^2 T_{\mathsf{N}}^2}\, df \\ &\overset{(a)}{=} \frac{2}{\pi^2 T_{\mathsf{N}}} \int_0^{\pi W_{\mathsf{N}}/2} \sin^2(\nu) \frac{1}{\pi T_{\mathsf{N}}}\, d\nu \\ &\overset{(4)}{=} \frac{2}{\pi^3 T_{\mathsf{N}}^2} \left[\frac{\pi W_{\mathsf{N}}}{4} - \frac{\sin(\pi W_{\mathsf{N}})}{4}\right] \\ &= \frac{W_{\mathsf{N}}}{2\pi^2 T_{\mathsf{N}}^2} \left[1 - \mathrm{sinc}(W_{\mathsf{N}})\right],\end{aligned} \tag{15}$$

and the denominator

$$\begin{aligned}\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} \mathcal{P}_{\mathsf{N}}(f)\, df &= 2T_{\mathsf{N}} \int_0^{B_{\mathsf{rx}}/2} \frac{\sin^2(\pi f T_{\mathsf{N}})}{\pi^2 f^2 T_{\mathsf{N}}^2}\, df \\ &\overset{(a)}{=} 2T_{\mathsf{N}} \int_0^{\pi W_{\mathsf{N}}/2} \frac{\sin^2(\nu)}{\nu^2} \frac{1}{\pi T_{\mathsf{N}}}\, d\nu \\ &\overset{(6)}{=} \frac{2}{\pi} \left[\mathrm{Si}(\pi W_{\mathsf{N}}) - \frac{2\sin^2(\pi W_{\mathsf{N}}/2)}{\pi W_{\mathsf{N}}}\right] \\ &= W_{\mathsf{N}} \left[\frac{2\mathrm{Si}(\pi W_{\mathsf{N}})}{\pi W_{\mathsf{N}}} - \mathrm{sinc}^2\left(W_{\mathsf{N}}/2\right)\right],\end{aligned} \tag{16}$$

where in both cases (a) accounts for the substitution $\nu = \pi f T_{\mathsf{N}}$. Plugging (15) and (16) in (3), we get

$$\beta_{\mathsf{N}}^2 = \frac{1}{2\pi^2 T_{\mathsf{N}}^2} \frac{1 - \mathrm{sinc}\left(W_{\mathsf{N}}\right)}{\left[2\mathrm{Si}\left(\pi W_{\mathsf{N}}\right)/(\pi W_{\mathsf{N}}) - \mathrm{sinc}^2\left(W_{\mathsf{N}}/2\right)\right]}, \tag{17}$$

[1] The sans-serif subscript N is used to emphasize that the related quantity is specific of the NRZ pulse – the same notation will be adopted for all of the other signals and related parameters, such as transmitted bandwidth.

and finally, using this result in (1), we obtain

$$\frac{\mathsf{MCRB}_{\mathsf{N}}(\tau)}{T_{\mathsf{N}}^2} = \frac{\left[\mathrm{Si}(\pi W_{\mathsf{N}}) + \frac{\cos(\pi W_{\mathsf{N}})-1}{\pi W_{\mathsf{N}}}\right]}{2\pi\,\mathsf{SNR}\left[W_{\mathsf{N}} - \frac{\sin(\pi W_{\mathsf{N}})}{\pi}\right]}. \tag{18}$$

IV. Sine-phased $\mathsf{BOC_s}(p,q)$

BOC signals with NRZ chips are commonly referred to as $\mathsf{BOC}(p,q)$ [4], where the first parameter p defines the sub-carrier rate $f_{\mathsf{sc}} \triangleq p f_0$, and the second parameter q defines the ranging code chiprate $R_{\mathsf{B}} \triangleq 1/T_{\mathsf{B}} = q f_0$. The ratio $n = 2f_{\mathsf{sc}}/R_{\mathsf{B}} = 2p/q$ represents the number of *half periods* of the square-wave sub-carrier during one code chip time, and can be either odd or even. There exist two versions of BOC signals: a sine-phased version $\mathsf{BOC_s}(p,q)$, in which the (odd-symmetric) square-wave subcarrier is obtained by hard-limitation of a sine waveform with the prescribed frequency, and a cosine-phased version, $\mathsf{BOC_c}(p,q)$, in which the (even-symmetric) square-wave subcarrier is obtained by hard-limitation of a cosine waveform.

The *sine-phased BOC* signal with sub-carrier frequency $f_{\mathsf{sc}} = 1/T_{\mathsf{sc}}$ is given by

$$x_{\mathsf{Bs}}(t) = \mathrm{sgn}\left[\sin(2\pi f_{\mathsf{sc}} t)\right] \sum_{k-\infty}^{\infty} c[k]\, p_{\mathsf{N}}(t - kT_{\mathsf{B}}), \tag{19}$$

and its PSD turns out to be [5]

$$\mathcal{P}_{\mathsf{Bs}}(f,p,q) = \begin{cases} T_{\mathsf{B}} \frac{\sin^2(\pi f T_{\mathsf{B}})}{(\pi f T_{\mathsf{B}})^2} \tan^2\left(\frac{\pi f T_{\mathsf{B}}}{n}\right), & n \text{ even} \\ T_{\mathsf{B}} \frac{\cos^2(\pi f T_{\mathsf{B}})}{(\pi f T_{\mathsf{B}})^2} \tan^2\left(\frac{\pi f T_{\mathsf{B}}}{n}\right), & n \text{ odd.} \end{cases} \tag{20}$$

To compute (3), let us recall the definition $W_{\mathsf{B}} = B_{\mathsf{rx}} T_{\mathsf{B}}$, and let us use (20) to compute separately numerator

$$\begin{aligned}&\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} f^2 \mathcal{P}_{\mathsf{Bs}}(f)\, df \overset{(b)}{=} \\ &= \frac{2n}{\pi^3 T_{\mathsf{B}}^2} \left[\frac{\varphi_0 \pi W_{\mathsf{B}}}{2n} + \sum_{m=1}^{n} \varphi_m \sin\left(\frac{m\pi W_{\mathsf{B}}}{n}\right)\right],\end{aligned} \tag{21}$$

and denominator

$$\begin{aligned}&\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} \mathcal{P}_{\mathsf{Bs}}(f)\, df \overset{(b)}{=} \\ &= \frac{2n}{\pi} \left[\sum_{m=1}^{n} \xi_m \mathrm{Si}\left(\frac{m\pi W_{\mathsf{B}}}{n}\right) + \frac{2n}{\pi W_{\mathsf{B}}} \sum_{m=0}^{n} \psi_m \cos\left(\frac{m\pi W_{\mathsf{B}}}{n}\right)\right],\end{aligned} \tag{22}$$

where in both cases (b) accounts for the substitution $\nu = \pi f T_{\mathsf{B}}/n$, and with the vector coefficients $\boldsymbol{\varphi} = [\varphi_0, \varphi_1, \ldots, \varphi_n]$, $\boldsymbol{\xi} = [\xi_1, \ldots, \xi_n]$, and $\boldsymbol{\psi} = [\psi_0, \psi_1, \ldots, \psi_n]$ that can be obtained by partial integration, and are summarized in Table I for some values of n for the reader's convenience. Combining (21) and (22), in (3), we get

Table I: Coefficients used to compute (23).

n	1	2	3	4
φ	$[1/2, -1/4]$	$[3/2, -1, 1/8]$	$[5/2, -2, 1/2, -1/12]$	$[7/2, -3, 1, -1/3, 1/16]$
ξ	$[-, 1]$	$[-, 1, -1/2]$	$[-, 8/9, -8/9, 1/3]$	$[-, 3/4, -1, 3/4, -1/4]$
ψ	$[-1/2, 1/2]$	$[-3/8, 1/2, -1/8]$	$[-5/18, 4/9, -2/9, 1/18]$	$[-7/32, 3/8, -1/4, 1/8, -1/32]$
n	5		6	
φ	$[9/2, -4, 3/2, -2/3, 1/4, -1/20]$		$[11/2, -5, 2, -1, 1/2, -1/5, 1/24]$	
ξ	$[-, 16/25, -24/25, 24/25, -16/25, 1/5]$		$[-, 5/9, -8/9, 1, -8/9, 5/9, -1/6]$	
ψ	$[-9/50, 8/25, -6/25, 4/25, -2/25, 1/50]$		$[-11/72, 5/18, -2/9, 1/6, -1/9, 1/18, -1/72]$	

Table II: Coefficients used to compute (29).

n	1	2	3
φ	$[3/2, -2, 1/4]$	$[5/2, -2, -1, 2/3, -1/8]$	$[7/2, -2, -2, 2/3, 1/2, -2/5, 1/12]$
ξ	$[-, 2, -1]$	$[-, 1/2, 1, -3/2, 1/2]$	$[-, 2/9, 8/9, -2/3, -8/9, 10/9, -1/3]$
ψ	$[-3/2, 2, -1/2]$	$[-5/8, 1/2, 1/2, -1/2, 1/8]$	$[-7/18, 2/9, 4/9, -2/9, -2/9, 2/9, -1/18]$
n	4		5
φ	$[9/2, -2, -3, 2/3, 1, -2/5, -1/3, 2/7, -1/16]$		$[11/2, -2, -4, 2/3, 3/2, -2/5, -2/3, 2/7, 1/4, -2/9, 1/20]$
ξ	$[-, 1/8, 3/4, -3/8, -1, 5/8, 3/4, -7/8, 1/4]$		$[-, 2/25, 16/25, -6/25, -24/25, 2/5, 24/25, -14/25, -16/25, 18/25, -1/5]$
ψ	$[9/32, 1/8, 3/8, -1/8, -1/4, 1/8, 1/8, -1/8, 1/32]$		$[-11/50, 2/25, 8/25, -2/25, -6/25, 2/25, 4/25, -2/25, -2/25, 2/25, 1/50]$
n	6		
φ	$[13/2, -2, -5, 2/3, 2, -2/5, -1, 2/7, 1/2, -2/9, -1/5, 2/11, -1/24]$		
ξ	$[-, 1/18, 5/9, -1/6, -8/9, 5/18, 1, -7/18, -8/9, 1/2, 5/9, -11/18, 1/6]$		
ψ	$[-13/72, 1/18, 5/18, -1/18, -2/9, 1/18, 1/6, -1/18, -1/9, 1/18, 1/18, -1/18, 1/72]$		

$$\beta_{\mathsf{Bs}}^2 = \frac{1}{\pi^2 T_{\mathsf{B}}^2} \cdot \frac{\frac{\varphi_0 \pi W_{\mathsf{B}}}{2n} + \sum_{m=1}^{n} \varphi_m \sin\left(\frac{m\pi W_{\mathsf{B}}}{n}\right)}{\sum_{m=1}^{n} \xi_m \,\mathrm{Si}\left(\frac{m\pi W_{\mathsf{B}}}{n}\right) + \frac{2n}{\pi W_{\mathsf{B}}} \sum_{m=0}^{n} \psi_m \cos\left(\frac{m\pi W_{\mathsf{B}}}{n}\right)}, \tag{23}$$

which, using (1), leads to

$$\frac{\mathsf{MCRB}_{\mathsf{Bs}}(\tau)}{T_{\mathsf{B}}^2} = \frac{1}{8\ \mathsf{SNR}} \cdot \frac{\sum_{m=1}^{n} \xi_m \,\mathrm{Si}\left(\frac{m\pi W_{\mathsf{B}}}{n}\right) + \frac{2n}{\pi W_{\mathsf{B}}} \sum_{m=0}^{n} \psi_m \cos\left(\frac{m\pi W_{\mathsf{B}}}{n}\right)}{\frac{\varphi_0 \pi W_{\mathsf{B}}}{2n} + \sum_{m=1}^{n} \varphi_m \sin\left(\frac{m\pi W_{\mathsf{B}}}{n}\right)}, \tag{24}$$

where $W_{\mathsf{B}} = B_{\mathsf{rx}} T_{\mathsf{B}}$.

V. Cosine-phased $\mathsf{BOC_c}(p, q)$

The *cosine-phased BOC* signal with sub-carrier frequency $f_{\mathsf{sc}} = 1/T_{\mathsf{sc}}$ is given by

$$x_{\mathsf{Bc}}(t) = \mathrm{sgn}[\cos(2\pi f_{\mathsf{sc}} t)] \sum_{k=-\infty}^{\infty} c[k]\, p_{\mathsf{N}}(t - kT_{\mathsf{B}}), \tag{25}$$

and its PSD turns out to be [5]

$$\mathcal{P}_{\mathsf{Bc}}(f, p, q) = \begin{cases} T_{\mathsf{B}} \dfrac{\sin^2(\pi f T_{\mathsf{B}})}{(\pi f T_{\mathsf{B}})^2} \dfrac{\left[\cos\left(\frac{\pi f T_{\mathsf{B}}}{n}\right) - 1\right]^2}{\cos^2(\pi f T_{\mathsf{B}}/n)}, & n \text{ even}, \\ T_{\mathsf{B}} \dfrac{\cos^2(\pi f T_{\mathsf{B}})}{(\pi f T_{\mathsf{B}})^2} \dfrac{\left[\cos\left(\frac{\pi f T_{\mathsf{B}}}{n}\right) - 1\right]^2}{\cos^2\left(\frac{\pi f T_{\mathsf{B}}}{n}\right)}, & n \text{ odd}. \end{cases} \tag{26}$$

To compute (3), let us recall the definition $W_{\mathsf{B}} = B_{\mathsf{rx}} T_{\mathsf{B}}$, and let us use (26) to compute separately numerator

$$\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} f^2 \mathcal{P}_{\mathsf{Bc}}(f)\, df \overset{(b)}{=} = \frac{2n}{\pi^3 T_{\mathsf{B}}^2} \left[\frac{\varphi_0 \pi W_{\mathsf{B}}}{2n} + \sum_{m=1}^{2n} \varphi_m \sin\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right)\right], \tag{27}$$

and denominator

$$\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} \mathcal{P}_{\mathsf{Bc}}(f)\, df \overset{(b)}{=} \frac{2n}{\pi} \cdot \left[\sum_{m=1}^{2n} \xi_m \mathrm{Si}\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right) + \frac{2n}{\pi W_{\mathsf{B}}} \sum_{m=0}^{2n} \psi_m \cos\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right)\right], \tag{28}$$

where in both cases (b) accounts for the substitution $\nu = \pi f T_{\mathsf{B}}/n$, and with the vector coefficients $\boldsymbol{\varphi} = [\varphi_0, \varphi_1, \ldots, \varphi_{2n}]$, $\boldsymbol{\xi} = [\xi_1, \ldots, \xi_{2n}]$, and $\boldsymbol{\psi} = [\psi_0, \psi_1, \ldots, \psi_{2n}]$ that can be obtained by partial integration, and are summarized in Table II for some values of n for the reader's convenience. Combining (27) and (28), in (3), we get

$$\beta_{\mathsf{Bc}}^2 = \frac{1}{\pi^2 T_{\mathsf{B}}^2} \cdot \frac{\frac{\varphi_0 \pi W_{\mathsf{B}}}{2n} + \sum_{m=1}^{2n} \varphi_m \sin\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right)}{\sum_{m=1}^{2n} \xi_m \,\mathrm{Si}\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right) + \frac{2n}{\pi W_{\mathsf{B}}} \sum_{m=0}^{2n} \psi_m \cos\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right)}, \tag{29}$$

which, using (1), leads to

$$\frac{\mathsf{MCRB}_{\mathsf{Bc}}(\tau)}{T_{\mathsf{B}}^2} = \frac{1}{8\ \mathsf{SNR}} \cdot \frac{\sum_{m=1}^{2n} \xi_m \,\mathrm{Si}\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right) + \frac{2n}{\pi W_{\mathsf{B}}} \sum_{m=0}^{2n} \psi_m \cos\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right)}{\frac{\varphi_0 \pi W_{\mathsf{B}}}{2n} + \sum_{m=1}^{2n} \varphi_m \sin\left(\frac{m\pi W_{\mathsf{B}}}{2n}\right)}. \tag{30}$$

VI. SRRC

In this case, the resulting unit-power raised-cosine PSD is

$$\mathcal{P}_{\mathsf{S}}(f;T_{\mathsf{S}},\alpha)=\begin{cases}T_{\mathsf{S}} & |f|\le\frac{1-\alpha}{2T_{\mathsf{S}}}\\ T_{\mathsf{S}}\cos^2\left[\frac{\pi T_{\mathsf{S}}}{2\alpha}\left(|f|-\frac{1-\alpha}{2T_{\mathsf{S}}}\right)\right] & \frac{1-\alpha}{2T_{\mathsf{S}}}<|f|\le\frac{1+\alpha}{2T_{\mathsf{S}}}\\ 0 & \text{elsewhere.}\end{cases} \tag{31}$$

Assuming as usual that $W_{\mathsf{S}} = B_{\mathsf{rx}}T_{\mathsf{S}} \ge (1+\alpha)$ (i.e., getting the whole signal power with no losses), we get

$$\begin{aligned}
&\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} f^2\mathcal{P}_{\mathsf{S}}(f;T_{\mathsf{S}},\alpha)\,df\\
&=2T_{\mathsf{S}}\int_0^{\frac{1-\alpha}{2T_{\mathsf{S}}}} f^2\,df\\
&+2T_{\mathsf{S}}\int_{\frac{1-\alpha}{2T_{\mathsf{S}}}}^{\frac{1+\alpha}{2T_{\mathsf{S}}}} f^2\cos^2\left[\frac{\pi T_{\mathsf{S}}}{2\alpha}\left(f-\frac{1-\alpha}{2T_{\mathsf{S}}}\right)\right]df\\
&\overset{(c)}{=}\frac{(1-\alpha)^3}{12T_{\mathsf{S}}^2}\\
&+2T_{\mathsf{S}}\int_0^{\alpha/T_{\mathsf{S}}}\left(\mu+\frac{1-\alpha}{2T_{\mathsf{S}}}\right)^2\cos^2\left(\frac{\pi T_{\mathsf{S}}}{2\alpha}\mu\right)d\mu\\
&\overset{(d)}{=}\frac{(1-\alpha)^3}{12T_{\mathsf{S}}^2}+2T_{\mathsf{S}}\int_0^{\pi/2}\left(\frac{2\alpha}{\pi T_{\mathsf{S}}}\right)^2\nu^2\cos^2(\nu)\frac{2\alpha}{\pi T_{\mathsf{S}}}\,d\nu\\
&+2T_{\mathsf{S}}\frac{2(1-\alpha)}{2T_{\mathsf{S}}}\int_0^{\pi/2}\frac{2\alpha}{\pi T_{\mathsf{S}}}\nu\cos^2(\nu)\frac{2\alpha}{\pi T_{\mathsf{S}}}\,d\nu\\
&+2T_{\mathsf{S}}\left(\frac{1-\alpha}{2T_{\mathsf{S}}}\right)^2\int_0^{\pi/2}\cos^2(\nu)\frac{2\alpha}{\pi T_{\mathsf{S}}}\,d\nu\\
&\overset{(9)(8)(5)}{=}\frac{(1-\alpha)^3}{12T_{\mathsf{S}}^2}\\
&+\frac{16\alpha^3}{\pi^3T_{\mathsf{S}}^2}\left[\frac{\pi^3}{48}+\frac{1}{8}\left(\frac{\pi^2}{2}-1\right)\sin\pi+\frac{\pi}{8}\cos\pi\right]\\
&+\frac{8(1-\alpha)\alpha^2}{\pi^2T_{\mathsf{S}}^2}\left[\frac{\pi^2}{16}+\frac{\pi}{8}\sin\pi+\frac{1}{8}(\cos\pi-1)\right]\\
&+\frac{(1-\alpha)^2\alpha}{\pi T_{\mathsf{S}}^2}\left[\frac{\pi}{4}+\frac{1}{4}\sin\pi\right]=\frac{1+\alpha^2\left(3-24/\pi^2\right)}{12T_{\mathsf{S}}^2},
\end{aligned} \tag{32}$$

where (c) and (d) account for the substitutions $\mu = f-\frac{1-\alpha}{2T_{\mathsf{S}}}$, and $\nu=\frac{\pi T_{\mathsf{S}}}{2\alpha}\mu$, respectively. Since, by construction, the denominator (signal power) is equal to 1, β_{S}^2 is equal to (32), which leads to

$$\frac{\mathsf{MCRB}_{\mathsf{S}}(\tau)}{T_{\mathsf{S}}^2}=\frac{3}{2\pi^2\mathsf{SNR}}\cdot\frac{1}{1+\alpha^2\left(3-\frac{24}{\pi^2}\right)}. \tag{33}$$

It is easy to see that the SRRC MCRB is minimized for $\alpha=0$.

VII. SRRC signals with subcarriers

In this case, the resulting PSD can be expressed as

$$\begin{aligned}&\mathcal{P}_{\mathsf{BL}}(f;p,q,\alpha)=\\&\quad=\frac{1}{2}\left[\mathcal{P}_{\mathsf{S}}\left(f-pf_0;T_{\mathsf{BL}},\alpha\right)+\mathcal{P}_{\mathsf{S}}\left(f+pf_0;T_{\mathsf{BL}},\alpha\right)\right],\end{aligned} \tag{34}$$

where $\mathcal{P}_{\mathsf{S}}(f;T,\alpha)$ is defined in (31). The total bandlimited offset carrier (BLOC) bandwidth turns out to be

$$B_{\mathsf{RF}}=2f_0\left[p+q\,\frac{(1+\alpha)}{2}\right]. \tag{35}$$

To compute (1), let us exploit the spectrum symmetry of (34), and consider the hypothesis $W_{\mathsf{BL}}=B_{\mathsf{rx}}T_{\mathsf{BL}}\ge 2p/q+(1+\alpha)$, where $T_{\mathsf{BL}}=1/(qf_0)$, to compute separately numerator

$$\begin{aligned}
&\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} f^2\mathcal{P}_{\mathsf{BL}}(f;p,q,\alpha)\,df\\
&=2\int_0^{B_{\mathsf{rx}}/2} f^2\mathcal{P}_{\mathsf{BL}}(f;p,q,\alpha)\,df\\
&=2\int_{f_0\left[p-q\frac{(1+\alpha)}{2}\right]}^{f_0\left[p+q\frac{(1+\alpha)}{2}\right]} f^2\mathcal{P}_{\mathsf{BL}}(f;p,q,\alpha)\,df\\
&\overset{(34)}{=}\int_{-qf_0\frac{(1+\alpha)}{2}}^{qf_0\frac{(1+\alpha)}{2}}(\nu+pf_0)^2\mathcal{P}_{\mathsf{S}}(\nu;T_{\mathsf{BL}},\alpha)\,d\nu\\
&\overset{(e)}{=}\int_{-qf_0\frac{(1+\alpha)}{2}}^{qf_0\frac{(1+\alpha)}{2}}\nu^2\,\mathcal{P}_{\mathsf{S}}(\nu;T_{\mathsf{BL}},\alpha)\,d\nu\\
&+p^2f_0^2\int_{-qf_0\frac{(1+\alpha)}{2}}^{qf_0\frac{(1+\alpha)}{2}}\mathcal{P}_{\mathsf{S}}(\nu;T_{\mathsf{BL}},\alpha)\,d\nu\\
&\overset{(f)}{=}f_0^2\left\{p^2+q^2\left[\frac{1+\alpha^2\left(3-24/\pi^2\right)}{12}\right]\right\},
\end{aligned} \tag{36}$$

in which (e) omits the term due to the cross-product, as the integral of an even function in a symmetric interval is 0, and in which (f) exploits (32) and the unit power of $\mathcal{P}_{\mathsf{S}}(\nu;T_{\mathsf{BL}},\alpha)$ over the full signal bandwidth, and denominator

$$\begin{aligned}
&\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2}\mathcal{P}_{\mathsf{BL}}(f;p,q,\alpha)\,df\\
&=2\int_0^{B_{\mathsf{rx}}/2}\mathcal{P}_{\mathsf{BL}}(f;p,q,\alpha)\,df\\
&=2\int_{f_0\left[p-q\frac{(1+\alpha)}{2}\right]}^{f_0\left[p+q\frac{(1+\alpha)}{2}\right]}\mathcal{P}_{\mathsf{BL}}(f;p,q,\alpha)\,df\\
&=2\int_{-qf_0\frac{(1+\alpha)}{2}}^{qf_0\frac{(1+\alpha)}{2}}\mathcal{P}_{\mathsf{S}}(\nu;T_{\mathsf{BL}},\alpha)\,d\nu=1.
\end{aligned} \tag{37}$$

Hence, β_{BL}^2 is equal to (36). Considering that $T_{\mathsf{BL}}=1/(qf_0)$, this leads to

$$\frac{\mathsf{MCRB}_{\mathsf{BL}}(\tau)}{T_{\mathsf{BL}}^2}=\frac{1}{8\pi^2\,\mathsf{SNR}\left\{\left(\frac{p}{q}\right)^2+\left[1+\frac{\alpha^2}{4}\left(1-\frac{6}{\pi^2}\right)\right]\right\}}. \tag{38}$$

VIII. Gaussian chip pulse

The Gaussian pulse has the property that its Fourier Transform is also a Gaussian function (i.e., it has a Gaussian-shaped frequency spectrum with no sidelobes), and is maximally compact both in the time and frequency domain [6] – for this reason it is often considered in the design of communication formats. A unit-power Gaussian ranging signal has the form

$$x_{\mathsf{G}}(t) = \sum_{k=-\infty}^{\infty} c[k]\, p_{\mathsf{G}}(t - kT_{\mathsf{G}}; \gamma), \tag{39}$$

where

$$p_{\mathsf{G}}(t;\gamma) = \sqrt[4]{\frac{2}{\pi\gamma^2}} \cdot \exp\left\{ -\left[\frac{t}{\gamma T_{\mathsf{G}}}\right]^2 \right\} \tag{40}$$

and where the chip time/chip rate are as usual $T_{\mathsf{G}} = 1/R_{\mathsf{G}}$. In (39)-(40), γ is a numerical parameter ≥ 0 that regulates the time and spectrum properties of the signal. The PSD of the Gaussian ranging signal turns out to be

$$\mathcal{P}_{\mathsf{G}}(f;\gamma) = \gamma T_{\mathsf{G}} \sqrt{2\pi} \cdot \exp\left\{ -2\left(\pi\gamma f T_{\mathsf{G}}\right)^2 \right\}. \tag{41}$$

Let us assume that $B_{\mathsf{rx}} \geq B_{\mathsf{G}}$, where B_{G}, computed as in

$$B_{\mathsf{G}} = \frac{2\sqrt{\ln(10)}}{\pi} \cdot \frac{R_{\mathsf{G}}}{\gamma} \Rightarrow R_{\mathsf{G}} = \frac{\gamma\pi B_{\mathsf{G}}}{2\sqrt{\ln(10)}}, \tag{42}$$

corresponds to the -20-dB pulse bandwidth, that can easily be derived using (41). This assumption is equivalent to $W_{\mathsf{G}} = B_{\mathsf{rx}} T_{\mathsf{G}} \geq 2\sqrt{\ln(10)}/(\pi\gamma)$, considered throughout the paper.

To compute the numerator of (3), let us exploit the symmetry of (41), so that

$$\begin{aligned}
&\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} f^2 \mathcal{P}_{\mathsf{G}}(f;\gamma)\, df \\
&= 2\int_0^{B_{\mathsf{rx}}/2} f^2 \mathcal{P}_{\mathsf{G}}(f;\gamma)\, df \overset{(g)}{\approx} 2\int_0^{+\infty} f^2 \mathcal{P}_{\mathsf{G}}(f;\gamma)\, df \\
&= 2\int_0^{+\infty} \sqrt{2\pi}\, \gamma T_{\mathsf{G}} f^2 \exp\left\{ -2\left(\pi\gamma f T_{\mathsf{G}}\right)^2 \right\} df \\
&\overset{(h)}{=} \frac{1}{\sqrt{\pi}\left(\pi\gamma T_{\mathsf{G}}\right)^2} \int_0^{+\infty} \nu^2 \exp\left\{-\nu^2\right\} d\nu \\
&\overset{(10)}{=} \frac{1}{\left(2\pi\gamma T_{\mathsf{G}}\right)^2},
\end{aligned} \tag{43}$$

where (g) accounts for the fact that almost all bandwidth has been considered, based on the hypothesis $B_{\mathsf{rx}} \geq B_{\mathsf{G}}$, and (h) performs the substitution $\nu = \sqrt{2}\left(\pi\gamma f T_{\mathsf{G}}\right)$. Since, by construction, the denominator (signal power) is equal to 1, β_{G}^2 is equal to (43). Combining it with (1), we finally get

$$\frac{\mathsf{MCRB}_{\mathsf{G}}(\tau)}{T_{\mathsf{G}}^2} \approx \frac{\gamma^2}{2\mathsf{SNR}} \tag{44}$$

and we can see, as expected, that $\mathsf{MCRB}_{\mathsf{G}}(\tau)$ decreases as γ decreases.

IX. GMSK

Besides the Gaussian pulse illustrated in Sect. VIII, another digital modulation format, often considered when we need to trade off bandwidth occupancy and compactness of the signal amplitude in the time domain, is the minimum shift keying (MSK) format, also including its Gaussian-filtered variants, the GMSK modulation. The baseband (G)MSK signal is given by [6], [7]

$$x_{\mathsf{M}}(t) = \exp\left\{ 2\pi h \sum_{k=-\infty}^{\infty} c[k]\, q_{\mathsf{M}}(t - kT_{\mathsf{M}}) \right\}, \tag{45}$$

where $c[k]$ is the k-th chip pulse of the ranging code, and, as usual, $T_{\mathsf{M}} = 1/R_{\mathsf{M}}$ is the (G)MSK chip time (the inverse of the chiprate). The coefficient h is the modulation index, usually set to $h = \frac{1}{2}$, and $q_{\mathsf{M}}(t)$ is the phase response

$$q_{\mathsf{M}}(t) = \int_{-\infty}^{t} g_{\mathsf{M}}(\tau)\, d\tau, \tag{46}$$

where $g_{\mathsf{M}}(t)$ is the pulse obtained from filtering a T_{M}-wide rectangular pulse with a Gaussian (low-pass) filter with -3-dB normalized bandwidth ω_{M},[2] and conventionally normalized so that its power is $\frac{1}{2}$ (and thus $q_{\mathsf{M}}(+\infty) = \frac{1}{2}$). This format guarantees that the phase of $x_{\mathsf{M}}(t)$ is *continuous* across chip variations (and hence it belongs to the family of continuous-phase modulation (CPM) formats), thus ensuring a theoretical constant signal envelope. Note that, when $\omega_{\mathsf{M}} = +\infty$, the Gaussian filter degenates into an all-pass filter, and hence $x_{\mathsf{M}}(t)$ (45) is actually a pure MSK format. In what follows, we adopt $\omega_{\mathsf{M}} = 0.3$, to obtain a more compact spectrum.

The PSD $\mathcal{P}_{\mathsf{M}}(f)$ cannot be expressed in a closed form, and hence the MCRB-related quantities. However, there exist many numerical methods (e.g., [7]) to evaluate $\mathcal{P}_{\mathsf{M}}(f)$.

To obtain (1) for the (G)MSK signal, we can first evaluate the autocorrelation function $R_{\mathsf{M}}(\tau)$ as a function of the phase response $q_{\mathsf{M}}(t)$ and considering the chips of the ranging code as equiprobable, using [7, Eq. (17)]. We can then obtain $\mathcal{P}_{\mathsf{M}}(f)$ as the Fourier transform of $R_{\mathsf{M}}(\tau)$, and eventually obtain the GB (3) by numerically integrating $\mathcal{P}_{\mathsf{M}}(f)$, so as to obtain the values of (1).

X. OFDM

A unit-power, symmetric-spectrum OFDM signal is given by

$$\begin{aligned}
x_{\mathsf{O}}(t) = &\frac{1}{\sqrt{N_{\mathsf{sc}}}} \cdot \\
&\cdot \sum_{k=0}^{N_{\mathsf{sc}}-1} \sum_{\ell=-\infty}^{\infty} c_k[\ell]\, p_{\mathsf{sc}}(t - \ell T_{\mathsf{sc}}) e^{\frac{j2\pi t}{T_{\mathsf{sc}}}\left[k - \frac{N_{\mathsf{sc}}-1}{2}\right]},
\end{aligned} \tag{47}$$

where N_{sc} is the number of orthogonal active subcarriers, T_{sc} is the *subcarrier chip time*, $1/T_{\mathsf{sc}}$ is the spacing

[2] In the literature, this parameter is usually indicated with β. Here, we use ω_{M} to avoid conflicts of notation.

between adjacent orthogonal subcarriers, $c_k[\ell]$ is the chip value on subcarrier k (having baseband frequency $(k - N_{\mathsf{sc}}/2 - 1)/T_{\mathsf{sc}}$) at the subcarrier chip time ℓT_{sc}, and $p_{\mathsf{sc}}(t)$ (the subcarrier chip pulse) is an NRZ unit-amplitude pulse with duration T_{sc}. What we call here *subcarrier chiprate* $R_{\mathsf{sc}} = 1/T_{\mathsf{sc}}$ is not commensurate with the chiprate of the other signals presented so far: instead, it represents the rate of the diverse *sub-codes* that we find on any of the subcarriers. Assuming that the OFDM modulator is fed with a "master" ranging code $c[k']$, running at the (master) chiprate $R_{\mathsf{O}} = 1/T_{\mathsf{O}}$, then we have $R_{\mathsf{O}} = N_{\mathsf{sc}} R_{\mathsf{sc}}$ – this is the *master chiprate* that is commensurate to that of the other signals. The relation between the chip index k' of the master code $c[k']$ and the subcarrier/time indices k and ℓ of the subcarriers codes $c_k[\ell]$ is $k' = \ell \cdot N_{\mathsf{sc}} + k$, $0 \le k \le N_{\mathsf{sc}} - 1$. In (47), we also neglect for simplicity the presence of a possible cyclic prefix and/or a spectrum shaping filter.

As is known, the OFDM signal is not strictly bandlimited. However, when N_{sc} is large, its RF bandwidth is practically equal to $N_{\mathsf{sc}}/T_{\mathsf{sc}} = 1/T_{\mathsf{O}} = R_{\mathsf{O}}$ with a very fast roll-off of the spectrum. Its PSD is in fact given by

$$\begin{aligned}\mathcal{P}_{\mathsf{O}}(f) &= \frac{T_{\mathsf{sc}}}{N_{\mathsf{sc}}} \sum_{k=0}^{N_{\mathsf{sc}}-1} \operatorname{sinc}^2\left[\left(f - \frac{k}{T_{\mathsf{sc}}} + \frac{N_{\mathsf{sc}}-1}{2T_{\mathsf{sc}}}\right) T_{\mathsf{sc}}\right] \\ &= T_{\mathsf{O}} \sum_{k=0}^{N_{\mathsf{sc}}-1} \operatorname{sinc}^2\left[f(N_{\mathsf{sc}}T_{\mathsf{O}}) - (k - (N_{\mathsf{sc}}-1)/2)\right],\end{aligned} \tag{48}$$

To obtain (1) for the OFDM signal (47), we consider the spectrum (48), and assume that $B_{\mathsf{rx}} \ge \frac{1}{T_{\mathsf{O}}} = \frac{N_{\mathsf{sc}}}{T_{\mathsf{sc}}}$. The denominator in the definition (3) of the (squared) GB is simple to evaluate

$$\begin{aligned}&\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} \mathcal{P}_{\mathsf{O}}(f)\, df \\ &= \frac{T_{\mathsf{sc}}}{N_{\mathsf{sc}}} \sum_{k=0}^{N_{\mathsf{sc}}-1} \int_{-N_{\mathsf{sc}}/(2T_{\mathsf{sc}})}^{N_{\mathsf{sc}}/(2T_{\mathsf{sc}})} \frac{\sin^2\left[\pi f\, T_{\mathsf{sc}} - \pi\left(k - \frac{N_{\mathsf{sc}}-1}{2}\right)\right]}{\left[\pi f\, T_{\mathsf{sc}} - \pi\left(k - \frac{N_{\mathsf{sc}}-1}{2}\right)\right]^2} df \\ &\overset{(e)}{=} \frac{1}{\pi N_{\mathsf{sc}}} \sum_{k=0}^{N_{\mathsf{sc}}-1} \int_{-\pi(k+1/2)}^{\pi[N_{\mathsf{sc}}-(k+1/2)]} \frac{\sin^2(\nu)}{\nu^2}\, d\nu \\ &\overset{(6)}{=} \frac{1}{\pi N_{\mathsf{sc}}} \sum_{k=0}^{N_{\mathsf{sc}}-1} \Bigg\{ \operatorname{Si}\left[\pi(2N_{\mathsf{sc}} - 2k - 1)\right] + \operatorname{Si}\left[\pi(2k+1)\right] \\ &\quad - \frac{1}{\pi(N_{\mathsf{sc}} - k - 1/2)} - \frac{1}{\pi(k+1/2)} \Bigg\},\end{aligned} \tag{49}$$

where (e) accounts for the substitution $\nu = \pi f T_{\mathsf{sc}} - \pi\left(k - \frac{N_{\mathsf{sc}}-1}{2}\right)$. We can easily see that, as long as $N_{\mathsf{sc}} \gg 1$, the result of this computation is equal to 1 irrespective of N_{sc}.

Computation of the numerator of (3) is more cumbersome:

$$\begin{aligned}&\int_{-B_{\mathsf{rx}}/2}^{B_{\mathsf{rx}}/2} f^2 \mathcal{P}_{\mathsf{O}}(f)\, df \\ &= \frac{T_{\mathsf{sc}}}{N_{\mathsf{sc}}} \sum_{k=0}^{N_{\mathsf{sc}}-1} \int_{-N_{\mathsf{sc}}/(2T_{\mathsf{sc}})}^{N_{\mathsf{sc}}/(2T_{\mathsf{sc}})} \frac{\sin^2\left[f\, T_{\mathsf{sc}} - \left(k - \frac{N_{\mathsf{sc}}-1}{2}\right)\right]}{\left[f\, T_{\mathsf{sc}} - \left(k - \frac{N_{\mathsf{sc}}-1}{2}\right)\right]^2} f^2 df \\ &\overset{(e)}{=} \frac{1}{\pi^3 N_{\mathsf{sc}} T_{\mathsf{sc}}^2} \cdot \\ &\cdot \sum_{k=0}^{N_{\mathsf{sc}}-1} \int_{-\pi(k+1/2)}^{\pi[N_{\mathsf{sc}}-(k+1/2)]} \frac{\sin^2(\nu)}{\nu^2} \left[\nu + \pi\left(k - \frac{N_{\mathsf{sc}}-1}{2}\right)\right]^2 d\nu \\ &= \frac{1}{\pi^3 N_{\mathsf{sc}} T_{\mathsf{sc}}^2} \sum_{k=0}^{N_{\mathsf{sc}}-1} \int_{-\pi(k+1/2)}^{\pi[N_{\mathsf{sc}}-(k+1/2)]} \sin^2(\nu)\, d\nu \\ &+ \frac{2}{\pi^2 N_{\mathsf{sc}} T_{\mathsf{sc}}^2} \cdot \\ &\cdot \sum_{k=0}^{N_{\mathsf{sc}}-1} \left(k - \frac{N_{\mathsf{sc}}-1}{2}\right) \int_{-\pi(k+1/2)}^{\pi[N_{\mathsf{sc}}-(k+1/2)]} \frac{\sin^2(\nu)}{\nu}\, d\nu \\ &+ \frac{1}{\pi N_{\mathsf{sc}} T_{\mathsf{sc}}^2} \cdot \\ &\cdot \sum_{k=0}^{N_{\mathsf{sc}}-1} \left(k - \frac{N_{\mathsf{sc}}-1}{2}\right)^2 \int_{-\pi(k+1/2)}^{\pi[N_{\mathsf{sc}}-(k+1/2)]} \frac{\sin^2(\nu)}{\nu^2}\, d\nu \\ &\overset{(4)(7)(6)}{=} \frac{N_{\mathsf{sc}}}{2\pi^2 T_{\mathsf{sc}}^2} + \frac{1}{\pi^2 N_{\mathsf{sc}} T_{\mathsf{sc}}^2} \cdot \\ &\sum_{k=0}^{N_{\mathsf{sc}}-1} \Bigg\{ \left(k - \frac{N_{\mathsf{sc}}-1}{2}\right) \Bigg[\log\left(-1 - \frac{N_{\mathsf{sc}}}{k+1/2}\right) \\ &\quad - \operatorname{Ci}\left(\pi(2N_{\mathsf{sc}} - 2k - 1)\right) + \operatorname{Ci}\left(\pi(2k+1)\right)\Bigg] \Bigg\} \\ &+ \frac{1}{\pi N_{\mathsf{sc}} T_{\mathsf{sc}}^2} \cdot \\ &\sum_{k=0}^{N_{\mathsf{sc}}-1} \Bigg\{ \left(k - \frac{N_{\mathsf{sc}}-1}{2}\right)^2 \Bigg[\operatorname{Si}\left[\pi(2N_{\mathsf{sc}} - 2k - 1)\right] \\ &\quad + \operatorname{Si}\left[\pi(2k+1)\right] - \frac{1}{\pi\left(N_{\mathsf{sc}} - k - \frac{1}{2}\right)} - \frac{1}{\pi\left(k + \frac{1}{2}\right)}\Bigg] \Bigg\},\end{aligned} \tag{50}$$

Just like for the case of the denominator, we see that, as long as $N_{\mathsf{sc}} \gg 1$, and according to [8], when the number of equispaced subcarriers is even and the spectrum is symmetric, we can get to the simplification

$$\beta_x^2 \simeq \frac{1}{T_{\mathsf{sc}}^2} \cdot \sum_{k=1}^{N_{\mathsf{sc}}/2} \left(k - \frac{1}{2}\right)^2 = \frac{1}{T_{\mathsf{sc}}^2} \cdot \frac{(N_{\mathsf{sc}}-1)^2}{12}, \tag{51}$$

which eventually leads to

$$\frac{\mathsf{MCRB}_{\mathsf{O}}(\tau)}{T_{\mathsf{sc}}^2} \simeq \frac{3}{2\pi^2} \frac{1}{\mathsf{SNR}(N_{\mathsf{sc}}-1)^2} \simeq \frac{3}{2\pi^2} \frac{1}{\mathsf{SNR}\ N_{\mathsf{sc}}^2}. \tag{52}$$

Recalling that $T_{\mathsf{sc}} = N_{\mathsf{sc}} T_{\mathsf{O}}$ we also get

$$\frac{\mathsf{MCRB}_{\mathsf{O}}(\tau)}{T_{\mathsf{O}}^2} \simeq \frac{3}{2\pi^2\ \mathsf{SNR}} = \left.\frac{\mathsf{MCRB}_{\mathsf{S}}(\tau)}{T_{\mathsf{S}}^2}\right|_{\alpha=0}. \tag{53}$$

The latter result is pretty natural, since in both cases (OFDM and SRRC with roll-off factor 0) the spectrum of the ranging signal is (approximately or exactly) *rectangular* with the same RF bandwidth.

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